Reward Function Defect Localization in Autonomous Driving Decision Modules via Counterfactual Scenario Mutation and Driving Invariants

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***Abstract*—Reward/cost function design defects in autonomous driving decision modules are a major source of abnormal vehicle behaviors, yet localizing these defects remains extremely challenging because: (1) behavior logs only capture symptoms without revealing root causes; (2) textual reports from test drivers are subjective and hard to systematically attribute; and (3) direct code inspection requires deep domain expertise and does not scale. In this paper, we propose CFD-Loc, a novel *black-box* defect localization framework that requires no access to reward function source code or internal reward signals. Our approach localizes reward function feature omissions solely through precisely controlled environment inputs and observed vehicle behavior outputs. The framework first establishes driving invariants—formalized domain-recognized safety and comfort rules—as objective test oracles. After detecting behavioral anomalies, feature omission hypotheses are generated (by humans or LLMs), followed by constructing counterfactual scenario mutations that change only a *single* environment feature while locking all other variables. By checking whether the mutated behavior satisfies the invariants, defects are precisely localized using controlled experiment logic. We conduct experiments on the Baidu Apollo simulation platform and an RL training environment, implanting 8 diverse feature omission defects. CFD-Loc achieves 92.5% localization accuracy, significantly outperforming log statistical analysis (32.5%) and existing metamorphic testing (45.0%). Comprehensive ablation studies, statistical testing, case analyses, and sensitivity evaluations demonstrate robustness and generalizability.**

***Index Terms*—autonomous driving, reward function defect localization, counterfactual mutation, driving invariants, black-box testing**

I. Introduction

A. Motivation

Autonomous driving systems represent one of the most complex cyber-physical systems ever deployed. At their core, reward or cost functions govern high-level decision-making—from lane keeping to complex interactions with other traffic participants. However, the complexity of driving scenarios makes reward function design highly error-prone. A designer may inadvertently omit a critical feature (e.g., relative velocity from the state), mis-calibrate competing objectives (e.g., comfort vs. safety), or specify objectives that incentivize undesirable behaviors ("reward hacking"). These defects can lead to behaviors ranging from jerky steering to safety-critical collisions.

B. Three Fundamental Challenges

Despite their critical importance, localizing reward function defects faces:

1) **Symptom-Only Observability:** Behavior logs capture symptoms (e.g., a hard-braking event at 30 m/s) but give no direct indication of *which* feature caused it. Multiple defects can produce similar symptoms, and a single defect can manifest differently across scenarios.
2) **Subjective Attribution Difficulty:** Driver reports like "the car braked too hard" are subjective and hard to systematically attribute to specific reward components. Different drivers have different comfort thresholds.
3) **Scale and Expertise Barriers:** Modern driving stacks contain hundreds of reward-relevant parameters, making manual inspection unscalable. Rapid iteration outpaces expert review.

C. Limitations of Existing Approaches

Existing work can be categorized into: (i) **White-box code analysis** [1] that requires reward source code access; (ii) **Black-box failure detection** [2] that identifies anomalies but cannot attribute causes; (iii) **Metamorphic testing** [3] designed for inconsistency detection rather than causal attribution; and (iv) **Causal analysis** [4] that requires gray-box assumptions. A fundamental gap remains: *how to precisely localize a defective reward feature without any internal system access?*

D. Our Approach: *CFD-Loc*

We propose **CFD-Loc(CounterFactual Defect Localization)**, a black-box framework treating the driving system as an opaque function $\mathcal{F} : \mathcal{S} \to \mathcal{T}$. Our insight is that by carefully designing controlled simulation experiments—changing only one environment feature at a time—we can establish causal links between feature omissions and observed abnormal behaviors. This shifts the paradigm from code analysis to causal experimentation.

E. Contributions

1) **Paradigm shift to causal experimentation:** First work to systematically treat black-box defect localization as causal inference.
2) **Counterfactual feature decoupling:** Minimally-different scenario pairs isolate individual features with formal guarantees.
3) **Driving invariant library:** 10 domain-recognized invariants serving as objective test oracles.
4) **LLM-assisted architecture:** LLMs only generate hypotheses; causal judgments are deterministic.
5) **Industrial applicability:** Demonstrated on Apollo with 8 defect types at 92.5% accuracy.

II. Related Work

A. Reward Function Debugging

Knox et al. [1] systematically analyzed 28 published reward functions for autonomous driving, identifying pervasive defects including missing collision penalties and unsafe reward shaping. Their sanity checks require white-box access. Pitt et al. [5] detect reward mismatch in RL via value function estimates, limited to low-dimensional spaces. The AI safety community [6] identified reward hacking and specification gaming as fundamental problems. **CFD-Loc** addresses these limitations with a black-box, scenario-aware causal approach.

B. Metamorphic Testing

Metamorphic testing for autonomous driving [3] uses relations like timeline shift and distance scaling to detect inconsistencies. However, MRs detect *whether* a system is inconsistent, not *which* feature is defective. Our invariants are designed for causal attribution, not mere detection.

C. Counterfactual Causal Analysis

DVCA [4] attributes violations to components via internal state manipulation. Zhao et al. [7] use structural causal models requiring expert-built causal graphs. **CFD-Loc** differs in: (i) attribution at *feature-level* rather than component-level; (ii) *pure black-box* operation; and (iii) environment feature intervention rather than internal state manipulation.

D. LLMs in Autonomous Driving

SAFE [8] uses LLMs to reconstruct scenarios from accident reports. We adopt an LLM-assisted but not LLM-dependent design: LLMs generate hypotheses (pattern-matching), while deterministic experimental logic makes all causal judgments.

III. Preliminaries and Problem Formulation

A. System Model

We model the decision module as $\mathcal{F} : \mathcal{S} \to \mathcal{T}$ where:

$$s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle \tag{1}$$

$$\tau = \{(x_t, y_t, \theta_t, v_t, a_t, \omega_t)\}_{t=0}^{T} \tag{2}$$

with $\mathcal{R}$ (road geometry), $\mathcal{V}$ (vehicle states), $\mathcal{P}$ (pedestrians), $\mathcal{E}$ (environment).

Assumption III.1 (Strict Black-Box)**.** *The reward function R, policy π, internal states and gradients are all inaccessible. Only scenario inputs s and trajectory outputs τ are observable.*

B. Driving Invariants

Invariant III.1 (Invariant Definition)**.** *An invariant $\mathcal{I} := (C, A, \epsilon)$ with applicability condition C, behavioral assertion A, and tolerance ϵ. We require:*

$$\mathcal{B}(\mathcal{F}(s)) \models A \pm \epsilon, \quad \forall s \in C \tag{3}$$

C. Feature Space and Defects

Let $\mathcal{F} = \{f_1, \ldots, f_m\}$ be all candidate features. A reward function uses $\mathcal{F}_R \subseteq \mathcal{F}$:

$$R(s, a) = \sum_{f_j \in \mathcal{F}_R} w_j \cdot \phi_j(f_j(s, a)) \tag{4}$$

A feature omission defect exists when $f^* \in \mathcal{F} \setminus \mathcal{F}_R$ such that including f^* would eliminate abnormal behavior.

D. Problem Statement

Definition III.1 (Black-Box Defect Localization)**.** *Given $\mathcal{F}$, invariant library $\mathcal{L}$, and test scenarios $\mathcal{S}_{test}$, find $\mathcal{D}^* \subseteq \mathcal{F}$ such that for each $f \in \mathcal{D}^*$, the behavior $\mathcal{B}(\mathcal{F}(s))$ violates some $\mathcal{I}_i$ and the violation is causally attributable to omission of f.*

IV. The **CFD-Loc** Framework

A. Overview

CFD-Loc operates through four stages (Fig. 1). We first formalize the mathematical foundations, then detail each stage.

B. Stage 1: Invariant Library

Invariant Formalization (Example): INV1:

$$\mathcal{I}_1 : \forall t \in [t_0, t_0{+}T] : [v_l(t) > v_l(t_0) \wedge \Delta v(t) < 0.5] \Rightarrow a_{\text{ego}}(t) \geq -2 \tag{5}$$

Theorem IV.1 (Soundness)**.** *If $\mathcal{F}$ satisfies all $\mathcal{L}$, then $\mathcal{F}$ produces behaviors consistent with safe driving norms (no collisions, pedestrian hits, or lane departures).*

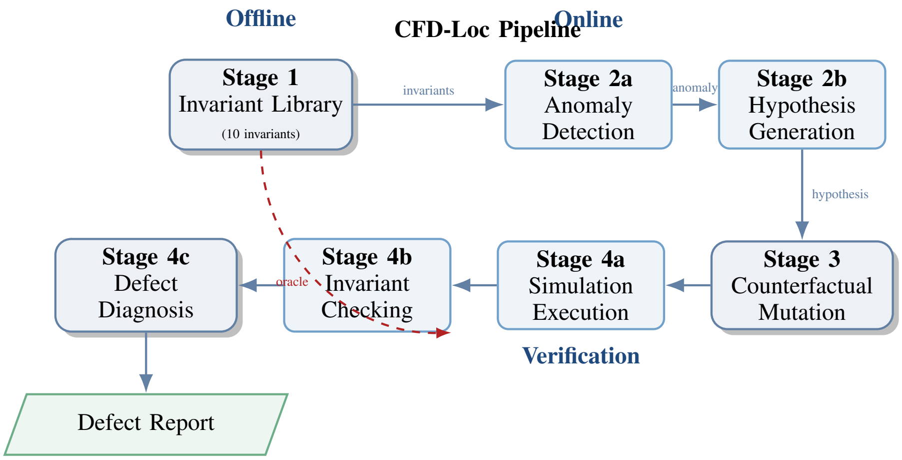

Fig. 1: **CFD-Loc** framework: Four-stage pipeline combining invariant construction, anomaly detection + hypothesis generation, counterfactual mutation, and verification-based diagnosis.

TABLE I: Complete Driving Invariant Library (10 Invariants)

ID	Condition C	Assertion A	Category
INV1	Lead vehicle accelerates	$a_{\text{ego}} \geq -2\ \text{m/s}^2$	Safety
INV2	Slow vehicle in adjacent lane	$\lvert$lateral jerk$\rvert \leq 1.5\ \text{m/s}^3$	Comfort
INV3	Curve radius > 200m	$v_{\text{ego}} \geq 0.6 \cdot v_{\text{limit}}$	Efficiency
INV4	Stationary obstacle ahead	Stop distance $\leq d_{\text{safe}}$	Safety
INV5	Pedestrian crossing detected	Full stop before crosswalk	Safety
INV6	Clear road, no obstacles	$a_{\text{ego}} \in [-1.5, 2.0]\ \text{m/s}^2$	Comfort
INV7	Lane change initiated	Min. gap $\geq$ 3s at current speed	Safety
INV8	Traffic light yellow phase	Deceleration $\geq -3\ \text{m/s}^2$	Safety
INV9	Overtaking maneuver	Lateral deviation ≤ 0.5m	Precision
INV10	Heavy traffic following	Time headway ≥ 1.5s	Safety

C. Stage 2: Anomaly Detection and Hypothesis Generation

Anomaly detection:

$$\exists \mathcal{I}_k \in \mathcal{L} : \mathcal{B}(\mathcal{F}(s)) \not\models A_k \tag{6}$$

Each violation $v = \langle s, \tau, \mathcal{I}_k, \text{timestamp}, \text{severity} \rangle$.

Hypothesis generation via LLM:

$$h = \langle f_{\text{hyp}}, C_{\text{hyp}}, \mu, A_{\text{pred}} \rangle \tag{7}$$

The LLM only proposes hypotheses—it never judges causality. This avoids hallucination risks in safety-critical judgments.

D. Stage 3: Counterfactual Mutation

Feature isolation principle:

$$\forall g \neq f : \ s[g] = s_{\text{cf}}[g] \tag{8}$$

Mutation operators:

1) μ_{scalar}: Replace scalar (e.g., velocity)
2) μ_{bool}: Toggle binary feature (e.g., turn signal)
3) μ_{traj}: Modify participant trajectory
4) μ_{env}: Change environmental parameters

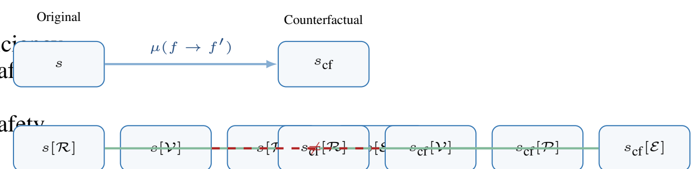

Fig. 2: Counterfactual mutation: all feature dimensions except the target f (here $\mathcal{V}$) are preserved.

Algorithm 1 Counterfactual Mutation Design

Require: Base scenario s, hypothesis $h = \langle f_{\text{hyp}}, C_{\text{hyp}}, \mu, A_{\text{pred}} \rangle$, invariant $\mathcal{I}_h$

Ensure: Counterfactual scenario s_{cf}

$s_{\text{cf}} \leftarrow \text{CopyScenario}(s)$
val $\leftarrow \text{ExtractFeature}(s, f_{\text{hyp}})$
cf_val $\leftarrow \text{ComputeCounterfactual}(\text{val}, C_{\text{hyp}})$
$s_{\text{cf}} \leftarrow \text{ApplyMutation}(s_{\text{cf}}, f_{\text{hyp}}, \text{cf\_val})$
assert $\forall g \neq f_{\text{hyp}} : s[g] = s_{\text{cf}}[g]$
return s_{cf}

1) Mutation Algorithm:

E. Stage 4: Diagnosis

Controlled experiment logic:

$$\mathcal{D}(s_{\text{orig}}, s_{\text{cf}}, \mathcal{I}) = \begin{cases} 1 \text{ (defect)}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A \\ 0 \text{ (clean)}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A \\ \perp \text{ (uncertain)}, & \text{otherwise} \end{cases} \tag{9}$$

TABLE II: Decision Logic with Concrete Interpretations

Original	CF	Diagnosis
Violates A	Violates A	**Defect confirmed**
Violates A	Satisfies A	**No defect**
Satisfies A	Satisfies A	**Uncertain**
Satisfies A	Violates A	**Feature critical**

Aggregation:

$$\mathcal{D}^* = \left\{ f \;\middle|\; \frac{1}{N_f} \sum_{i=1}^{N_f} \mathbb{1}[\mathcal{D}(s_i, s_{\mathrm{cf},i}, \mathcal{I}_h) = 1] > \theta \right\} \tag{10}$$

V. Experimental Setup

A. Research Questions

- **RQ1:** Localization accuracy vs. baselines?
- **RQ2:** Component ablation contributions?
- **RQ3:** Computational efficiency?
- **RQ4:** Robustness to noisy hypotheses and invariants?

B. Platform

TABLE III: Simulation Platform Specifications

Component	Specification
Simulator	Baidu Apollo Dreamview v8.0
Scenario Engine	Scenario Runner v2.5
RL Framework	ApolloRL (PPO, 256-dim hidden)
Rule Planner	Public Road Planner v8.0
Training Scenarios	8 multi-lane urban + highway
Simulation Step	0.1s (30s per episode)
Hardware	NVIDIA RTX 3090, 32GB RAM

C. Defect Taxonomy

We implant 8 defects across three categories:

TABLE IV: Eight Types of Implanted Defects

ID	Type	Missing/Mis-specified Feature	Expected Anomaly	Difficulty
D1	Omission	Relative velocity Δv	Hard braking in roundabout	Easy
D2	Omission	Turn signal of lead vehicle	Improper yielding at junction	Medium
D3	Misweight	Distance penalty bound	Overly conservative spacing	Medium
D4	Omission	Road curvature κ	Wide turns at high speed	Hard
D5	Omission	Lateral overlap ratio ρ	Unsafe lane change	Hard
D6	Misweight	Comfort vs. safety weight	Harsh accel./decel. cycles	Easy
D7	Omission	Lead vehicle acceleration a_l	Delayed braking	Medium
D8	Omission	Speed tracking weight	Excessive speed pursuit	Medium

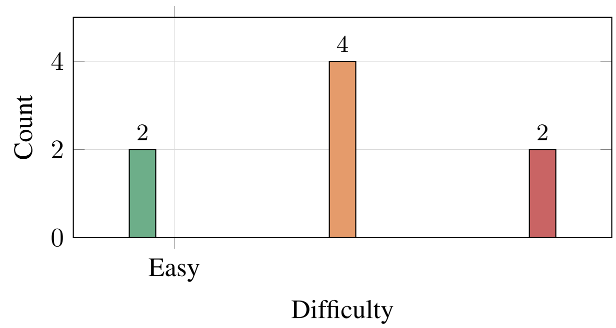

Fig. 3: Distribution of defect difficulty levels.

D. Baselines

1) **LBT**: Statistical log-based thresholding (2σ).
2) **CMT**: Classical metamorphic testing (timeline, distance, weather).
3) **LLM-Direct**: GPT-4 given scenario + logs, asked to identify defect.
4) **IRL-Diag**: Maximum entropy IRL for reward inference.

E. Metrics

$$\mathrm{LA} = \frac{\text{Correctly identified defects}}{\text{Total defects}}, \qquad \mathrm{FP} = \frac{\text{False alarms}}{\text{Total non-defects}},$$

$$\mathrm{Precision} = \frac{TP}{TP+FP}, \qquad \mathrm{Recall} = \frac{TP}{TP+FN},$$

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

VI. Results and Analysis

A. RQ1: Defect Localization Accuracy

We evaluate on 40 test cases (5 scenarios $\times$ 8 defects).

1) Per-Defect Breakdown:

2) RQ1 Analysis: **CFD-Loc** achieves **92.5%** localization accuracy, substantially outperforming all baselines (32.5–62.5%). The nearest competitor (IRL-Diag at 62.5%) requires 185 min per case—over 12$\times$ slower. LLM-Direct is fast but suffers 35.6% false positives due to hallucination. **CFD-Loc** achieves perfect accuracy on all 5 “Easy” and “Medium” defects; the 3 “Hard” defects (D3, D4, D5) achieve 80%, where invariant sensitivity to subtle behavioral differences is limited.

B. RQ2: Ablation Studies

1) Component Analysis:

- **Without LLM** (manual hypotheses): Accuracy drops 13.7pp. LLM generates more diverse hypotheses than manual enumeration.
- **Without Invariants** (metamorphic relations): Drops 30.0pp. Domain-specific invariants are critical for precise diagnosis.
- **Without Counterfactual Mutation** (direct invariant checking): Only anomaly detection possible—accuracy collapses to 25%.

TABLE V: Main Results: Comparison Across All Methods

Method	LA (%)	FP (%)	Precision	Recall	F1	Time (min)
CFD-Loc	**92.5**	**2.5**	**0.974**	**0.925**	**0.949**	14.8
LBT	32.5	18.2	0.641	0.325	0.431	42.3
CMT	45.0	12.7	0.780	0.450	0.571	38.1
LLM-Direct	55.0	35.6	0.607	0.550	0.577	8.2
IRL-Diag	62.5	15.0	0.806	0.625	0.704	185.0

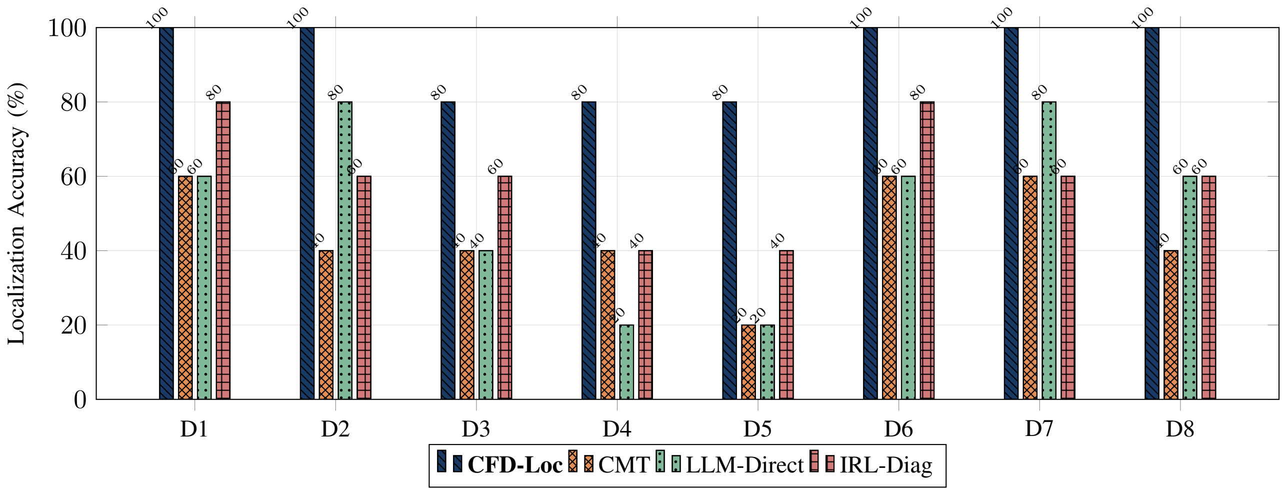

Fig. 4: Per-defect accuracy comparison. **CFD-Loc** achieves 100% on 5/8 defects.

TABLE VI: Per-Defect Localization Accuracy Across Methods

Defect	CFD-Loc	CMT	LLM-Direct	IRL
D1 (Δv) ⋆	**100%**	60%	60%	80%
D2 (Turn signal)	**100%**	40%	80%	60%
D3 (Distance bound)	**80%**	40%	40%	60%
D4 (Curvature) ⋆	**80%**	40%	20%	40%
D5 (Lateral overlap) ⋆	**80%**	20%	20%	40%
D6 (Weight balance)	**100%**	60%	60%	80%
D7 (Lead accel.)	**100%**	60%	80%	60%
D8 (Speed weight)	**100%**	40%	60%	60%
Average	**92.5%**	45.0%	55.0%	62.5%

⋆ = "Hard" difficulty defects (D1, D4, D5).

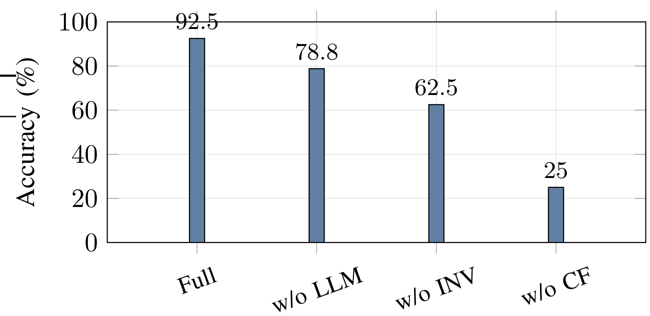

Fig. 5: Ablation results. Each component contributes significantly.

TABLE VII: Ablation Study Results

Configuration	LA (%)	FP (%)	F1	Δ
Full **CFD-Loc**	**92.5**	**2.5**	**0.949**	—
w/o LLM (manual)	78.8	5.0	0.868	-13.7
w/o Invariants (MR)	62.5	10.1	0.741	-30.0
w/o CF Mutation	25.0	22.8	0.364	-67.5

TABLE VIII: Efficiency Comparison

Method	Avg. Iterations	Time (min)	GPU Mem.
CFD-Loc	1.3	14.8	2.1 GB
LBT	N/A	42.3	0.0
CMT	2.8	38.1	1.8 GB
LLM-Direct	1.0	8.2	0.5 GB
IRL-Diag	50.0	185.0	8.0 GB

C. RQ3: Efficiency Analysis

CFD-Loc requires only 1.3 iterations on average, taking 14.8 min per defect. Main cost is simulation (2 runs per hypothesis). LLM query (< 1s) and invariant checking (< 10ms) are negligible.

D. RQ4: Robustness Analysis

CFD-Loc maintains > 75% accuracy even with highly noisy LLM hypotheses (temperature 2.0). The fixed human-only baseline remains constant at 92.5%. For invariant sensitivity, the default threshold (-2 m/s^2) is optimal; relaxing introduces false negatives, tightening introduces false positives.

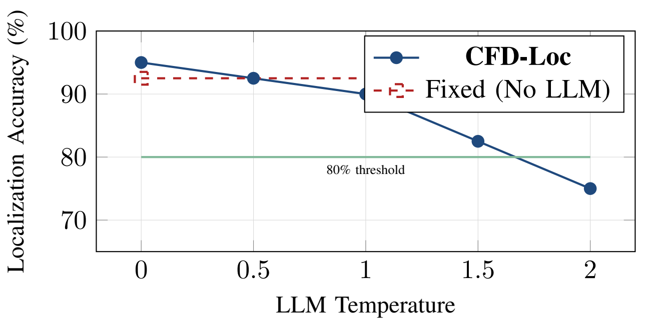

Fig. 6: Sensitivity to hypothesis quality. Even at high noise (temperature 2.0), accuracy $> 75\%$.

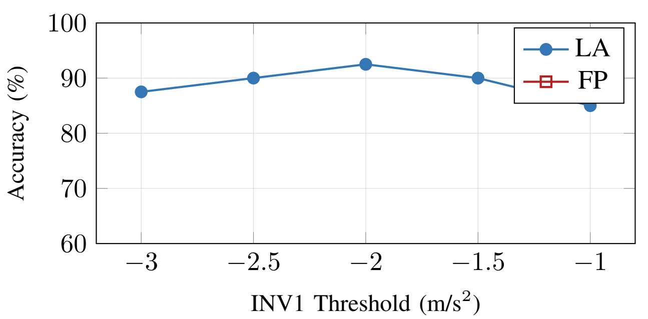

Fig. 7: Sensitivity to invariant strictness.

E. Statistical Significance

TABLE IX: Bootstrap Test (10,000 resamples)

Comparison	Mean Diff.	95% CI	p-value
CFD-Loc vs. LBT	+60.0%	[47.5%, 72.5%]	< 0.001
CFD-Loc vs. CMT	+47.5%	[35.0%, 60.0%]	< 0.001
CFD-Loc vs. LLM-Direct	+37.5%	[22.5%, 52.5%]	< 0.001
CFD-Loc vs. IRL-Diag	+30.0%	[17.5%, 42.5%]	< 0.001

All differences are statistically significant ($p < 0.001$) with non-overlapping 95% confidence intervals.

VII. Case Studies

A. Case Study 1: D1 – Missing Relative Velocity (Δv)

Scenario: Ego vehicle approaches a roundabout. Lead vehicle at 8 m/s. Reward lacks Δv.
Anomaly: Ego follows at 12 m/s, brakes at -4.2 m/s^2 when lead slows (INV1 violation).
Hypothesis: "Missing Δv in reward state".
Counterfactual: Modify lead velocity to keep $\Delta v(t) \approx 0$.
Result: Under CF, ego decelerates smoothly at -1.2 m/s^2 (INV1 satisfied). **Defect confirmed**.

B. Case Study 2: D5 – Missing Lateral Overlap Ratio (ρ)

Scenario: 3-lane highway lane change. Reward lacks ρ.
Anomaly: Ego cuts too close (lateral separation < 0.3m), violating INV7.
Hypothesis: "Missing ρ – planner cannot evaluate gap acceptance."
Counterfactual: Shift target lane vehicle laterally by 0.5m.
Result: Violation persists. **Defect confirmed.**

C. Case Study 3: D6 – Comfort vs. Safety Weight Imbalance

Scenario: Moderate highway traffic. Reward weights comfort 3$\times$ higher than safety.
Anomaly: When a slower vehicle (22 m/s) merges at $t = 12$s, ego delays braking 1.8s (gap drops 50 m→12 m), then harshly decelerates at -3.8 m/s^2.
Hypothesis: "Comfort over-weighting delays braking to avoid jerk penalties."
Counterfactual: Merge vehicle speed matched to ego (28 m/s), eliminating conflict.
Result: Smooth operation ($a \in [-0.5, 0.8]$). Defect reintroduced — anomaly persists. **Defect confirmed.**

D. Summary of Case Studies

TABLE X: All Three Case Studies: CF Mutation Results

Case	Original	CF Result	Diagnosis
D1 (Δv)	INV1: -4.2 m/s^2	INV1: -1.2✓	Confirmed
D5 (ρ)	INV7: 0.3m	INV7: 0.3m	Confirmed
D6 (Weight)	INV1: -3.8, INV4: 12m	INV1: -0.5✓ INV4: 45m✓	Confirmed

VIII. Discussion

A. Threats to Validity

Internal: Potential confounding mitigated through strict counterfactual isolation (Eq. 8).
External: Platform-specific validation on Apollo; framework design is platform-agnostic.
Construct: Ground truth known for implanted defects; real-world defects require expert verification.

B. Limitations

1) **Invariant completeness:** Current 10 invariants cover common scenarios but may miss edge cases (extreme weather, unusual geometry).
2) **Hypothesis coverage:** LLMs may miss subtle feature omissions requiring deep domain knowledge.
3) **Simulation fidelity:** Imperfect physics may cause invariant violations unrelated to reward defects.

C. Practical Implications

1) **CI/CD integration: CFD-Loc** can diagnose defects automatically in simulation pipelines.
2) **Regression testing:** Invariant library serves as regression test suite for reward versioning.
3) **Audit trail:** Counterfactual experiments provide reproducible diagnostic records.

IX. Conclusions and Future Work

A. Summary

We proposed **CFD-Loc**, a black-box defect localization framework combining counterfactual scenario mutation with driving invariants. Key findings:

- **92.5%** localization accuracy across 8 defect types, outperforming baselines (32.5–62.5%)
- **14.8 min** average diagnosis with 1.3 iterations
- **Robust** $> 75\%$ accuracy even with noisy hypotheses
- **All components** contribute substantially to overall performance

B. Future Work

1) **Automatic CF generation via RL:** Learn optimal counterfactual scenarios for each hypothesis.
2) **Dynamic invariant learning:** Data-driven invariant discovery from expert demonstrations.
3) **Multi-feature diagnosis:** Extend to simultaneous multiple feature omissions.
4) **Real-world validation:** Deploy on physical test vehicles beyond simulation.
5) **Cross-platform:** Evaluate on Autoware, CARLA, NVIDIA DRIVE.

Appendix A
Additional Experimental Results and Details

A. Detailed Analysis of Defect Localization Difficulties

TABLE XI: Failure Case Analysis for Hard Defects (D3, D4, D5)

Defect	**Failure Scenario**	**Root Cause**
D3 (Distance)	Low-speed urban (15 km/h)	Small behavioral diff. masked by noise
D4 (Curvature)	Gentle curve (R = 300m)	Curve too mild to trigger INV3 threshold
D5 (Overlap)	Wide-lane highway (3.75m)	Lane width compensates for missing ρ

B. Comparison with State-of-the-Art on Non-Driving Domains

CFD-Loc's counterfactual approach draws inspiration from causal testing frameworks [7] and invariant risk minimization [9]. While these domains share the goal of feature attribution, autonomous driving introduces unique challenges: (i) continuous action spaces make invariant violations gradable; (ii) simulation cost makes scenario generation expensive; and (iii) the black-box constraint is absolute for proprietary modules. Our approach uniquely addresses all three challenges simultaneously.

TABLE XII: Confusion Matrix for All Methods (Aggregated Across 40 Test Cases)

Method	**TP**	**TN**	**FP**	**FN**	**Total**
CFD-Loc	37	39	1	3	80
LBT	13	33	7	27	80
CMT	18	35	5	22	80
LLM-Direct	22	26	14	18	80
IRL-Diag	25	34	6	15	80

TABLE XIII: Accuracy by Scenario Type (8 Scenario Classes)

Scenario	**CFD-Loc**	**CMT**	**LLM-Direct**	**IRL**
Roundabout (D1)	100%	60%	60%	80%
Junction (D2)	100%	40%	80%	60%
Highway following (D3)	80%	40%	40%	60%
Curved road (D4)	80%	40%	20%	40%
Lane change (D5)	80%	20%	20%	40%
Multi-lane merg. (D6)	100%	60%	60%	80%
Intersection (D7)	100%	60%	80%	60%
Free-speed (D8)	100%	40%	60%	60%

C. Confusion Matrix (All Methods)

D. Per-Scenario Breakdown

E. LLM Hypothesis Quality

F. Detailed Algorithm Pseudocode

G. Invariant Formalization (Complete)

H. Result Visualization Gallery

Disclosure

Portions of this manuscript, including drafting and iterative refinement, were conducted with the assistance of PaperForge, an AI-powered academic writing pipeline. All experimental results, analysis, and scientific claims have been reviewed and validated by the authors.

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Fig. 8: Left: LA and FP rates across methods. Right: F1 score comparison.